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role in the treatment. The disease has been treated through diet, physical exercises, and administration of medications.³

Genicular nerve ablation has gained prominence. It is a procedure to interrupt the pain signaling from the knee to the brain by destroying the nerves. Ablation can be performed using different methods, including the application of phenol and radiofrequency. Both methods effectively reduce chronic knee pain and improve the function and quality of life of patients with knee osteoarthritis who are not candidates for or have failed conservative treatment.⁴

Phenol is a well-established neurolytic chemical agent, acting through a demyelinating and possibly ischemic effect.⁵⁻⁶ Its use does not require expensive equipment, and genicular ablation can be performed with few resources.⁷ On the other hand, radiofrequency ablation works by transforming energy into heat, destroying neural tissue.⁸

The aim of this randomized study is to analyze the comparative efficacy of genicular nerve ablation using phenol or radiofrequency in patients with severe knee osteoarthritis. By investigating functional outcomes, we hope to provide substrates that can guide clinical practice and improve pain management in patients with this debilitating condition.

MATERIALS AND METHODS

This study was conducted by the Osteometabolic Diseases Group of the Department of Orthopedics and Traumatology at the Hospital das Clínicas, University of São Paulo. Patients with knee osteoarthritis were selected and randomized into 2 groups to undergo genicular nerve ablation using radiofrequency (group 1) or phenol (group 2). The study was approved by the University Ethics Committee, under protocol number 1574. The Informed Consent Form was signed by all study participants.

Inclusion criteria used: men and women diagnosed with severe unilateral or bilateral knee osteoarthritis, with a formal indication for unilateral or bilateral total knee arthroplasty, who had a pain score greater than 6 on the Numerical Pain Scale (NPS) and a positive response to the genicular nerve anesthetic block test, in addition to the ability to read, understand, and respond to the questionnaires. The exclusion criteria were: (1) patients who had previously undergone knee arthroplasty on the studied knee, (2) use of a pacemaker, (3) patients who had undergone electroacupuncture, (4) patients who had knee injections in the last 6 months, (5) patients with cognitive, psychiatric, or neurological disorders that interfere with attention, memory, logical reasoning, or comprehension functions, thus preventing them from assimilating the given instructions, (6) use of anticoagulants between the genicular nerve block test and the intervention, (7) recent knee pain of less than 3 months, (8) advanced liver disease, and (9) signs of systemic or local infection before the intervention.

Patients were evaluated regarding personal characteristics and knee functionality scores. The ablation procedures were performed between December 2022 and January 2023, and the patients were followed up after this period.

The personal characteristics of the individuals evaluated were: age, sex, weight, height, and BMI.

Patients were asked to rate their pain using the Numerical Pain Scale (NPS) and were subjected to the functional tests Timed Up and

The quantitative personal characteristics or initial scores evaluated were described according to groups using summary measures (means, standard deviations, medians, and quartiles) and compared between groups using unpaired t-Student tests. The sex of the patients was described according to groups using absolute and relative frequencies, and the association was verified using Fisher's exact test. The questionnaire scores were described according to groups over the evaluated time points using summary measures and compared between groups and time points using generalized estimating equations (GEE) with normal marginal distribution and identity link function or logarithmic link function, assuming an autoregressive correlation matrix of the first order (AR(1)) between the evaluation moments for all analyses. The analyses were followed by Bonferroni multiple comparisons to verify where the differences occurred between the groups and evaluation moments. The analyses were performed using IBM-SPSS for Windows version 22.0 and tabulated using Microsoft-Excel 2013 software, with tests conducted at a significance level of 5%.

RESULTS

During the period from December 2022 to January 2023, 35 patients were treated in a blinded and randomized manner with the genicular nerve ablation procedure using either radiofrequency (study group – 17 individuals) or phenol (control group – 18 individuals), adhering to the study's inclusion and exclusion criteria.

Table 1 shows that the personal characteristics of the patients were similar between the groups, indicating that the groups are comparable.

The patients were evaluated at a pre-intervention/initial time point regarding the NRS, KOOS score, and physical performance tests (TUG and 30s-STs). The KOOS score was evaluated separately in its 5 pillars (Symptoms + Stiffness, Pain, Daily Activities, Sports and Leisure Activities, and Quality of Life) as well as overall, with no significant differences found between the groups at the initial time point, as recorded in Table 2. The initial NRS recorded was also not different between the groups. Therefore, the groups presented similar symptomatology and activity levels at the beginning of the study, making them comparable.

Table 1. Description of personal characteristics by groups and results of statistical tests.

Variable	Group		Total (N = 35)	p
	Study (N = 17)	Control (N = 18)		
Age (years)				0.430**
Average SD	63.9 ± 8.8	61.3 ± 10	62.6 ± 9.4	
median (p25; p75)	66 (58; 69.5)	60 (53; 69.3)	65 (56; 69)	
Gender, n (%)				0.146
Female	10 (58.8)	15 (83.3)	25 (71.4)	
Male	7 (41.2)	3 (16.7)	10 (28.6)	
Weight (kg)				0.267**
Average ± SD	92.3 ± 17.4	85.7 ± 17.3	88.9 ± 17.4	
median (p25; p75)	91.9 (79.3; 107.1)	81.8 (72.8; 93.8)	86 (75.5; 96.2)	
Hight (cm)				0.414**
Average SD	162 ± 10.9	159.3 ± 8	160.6 ± 9.5	
median (p25; p75)	162 (153; 169.5)	159 (154.5; 163.5)	160 (153; 166)	
BMI (Kg/m)				0.531**

		7) Control (N = 18)		
				0.253**
		9.1 ± 1	9.2 ± 0.9	
median (p25; p75)	10 (9; 10)	9 (8; 10)	10 (8; 10)	
TUG				0.949**
Average ± SD	17.5 ± 9.3	17.3 ± 8.2	17.4 ± 8.6	
median (p25; p75)	14 (11.8; 17.4)	14.3 (11.6; 18.7)	14.3 (11.8; 17.2)	
TLS30S				0.343**
Average ± SD	7 ± 2.3	6.3 ± 1.9	6.6 ± 2.1	
median (p25; p75)	7 (5; 9)	6 (5; 8)	7 (5; 8)	
KOOS General				0.769**
Average ± SD	25.8 ± 17	24.3 ± 12.5	25 ± 14.6	
median (p25; p75)	24 (10.5; 41)	27 (10; 35)	26 (10; 37)	
KOOS Symptoms+Rigidity				0.318£
Average ± SD	38.7 ± 26.7	29.8 ± 13.1	34.1 ± 21	
median (p25; p75)	50 (9; 61)	32 (20.3; 40.8)	32 (14; 54)	
KOOS Pain				0.684£
Average ± SD	24.9 ± 16.3	27.4 ± 16.6	26.2 ± 16.2	
median (p25; p75)	28 (8; 40.5)	34.5 (7.5; 39.8)	33 (8; 39)	
KOOS Daily Activity				0.935£
Average ± SD	29.1 ± 20.8	29 ± 17.7	29.1 ± 19	
median (p25; p75)	28 (11; 47)	33.5 (11.5; 41.8)	32 (12; 44)	
KOOS Sports and Leisure Activity				0.613£
Average ± SD	6.8 ± 9.2	4.2 ± 5.8	5.4 ± 7.6	
median (p25; p75)	0 (0; 10)	0 (0; 6.3)	0 (0; 10)	
KOOS Quality of Life				0.351£
Average ± SD	13.4 ± 10.6	10.6 ± 12.8	11.9 ± 11.7	
median (p25; p75)	13 (0; 22)	6 (0; 13.5)	13 (0; 19)	
Adverse reaction. n (%)				0.697
No	4 (36.4)	6 (46.2)	10 (41.7)	
Yes	7 (63.6)	7 (53.8)	14 (58.3)	
Have used any analgesic?. n (%)				>0.999
No	7 (70)	8 (66.7)	15 (68.2)	
Yes	3 (30)	4 (33.3)	7 (31.8)	
Increased use of analgesics?. n (%)				0.495#
Decreased	8 (80)	8 (66.7)	16 (72.7)	
Unchanged	2 (20)	3 (25)	5 (22.7)	
Increased	0 (0)	1 (8.3)	1 (4.5)	
Was satisfied?. n (%)				0.411#
Unsatisfied	0 (0)	1 (8.3)	1 (4.5)	
Indifferent	1 (10)	2 (16.7)	3 (13.6)	
Satisfied	8 (80)	6 (50)	14 (63.6)	
Very satisfied	1 (10)	3 (25)	4 (18.2)	

Exact Fisher test; # Test of the likelihood ratio; ** Unpairing t-Student test; £ Mann-Whitney test

Finally, the degree of patient satisfaction, the presence of adverse reactions, and the use of analgesics were also recorded. In our study, complications were mainly observed between the first and third weeks, including: eight cases of pain at the puncture site, eight cases of local edema, seven cases of increased local temperature, eight cases of ecchymosis, four cases of paresthesia, and one case of local bleeding. No serious adverse events were observed.

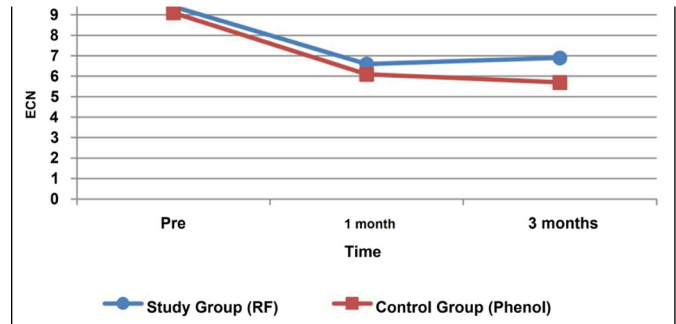


Figure 1. Graph with results of pain assessment using the numeric rating scale (NRS) at the 3 evaluated time points.

Table 3 shows that the average behavior of the patients was the same between the groups over the evaluation periods for all assessed scores (Interaction $p > 0.05$), none of the scores differed on average between the groups regardless of the evaluation periods (Group $p > 0.05$), and almost all scores showed significant average differences over the evaluation periods (Moment $p < 0.05$), with the exception of KOOS sports and leisure activities and KOOS quality of life (Moment $p = 0.052$ and Moment $p = 0.053$, respectively). These findings demonstrate the positive results of the interventions performed, with no significant differences between them (radiofrequency vs. phenol).

Table 4 shows that all scores that differed over the evaluation periods represented clinically relevant improvements associated with the treatments performed. This is demonstrated by the significant average improvement from the pre/initial period to the other evaluated moments ($p < 0.05$). The TUG, in turn, showed a significant average improvement between the initial evaluation and the 1-month evaluation ($p = 0.013$). These findings corroborate those recorded in Table 3, demonstrating that both interventions were effective in controlling pain and improving quality of life.

DISCUSSION

Both methods of genicular nerve ablation proved effective for symptom control and improvement in the quality of life of patients with severe knee osteoarthritis. Comparative analysis showed that there were no significant differences between the groups in terms of knee functionality scores over the evaluation periods (Table 3). The improvement showed a lasting effect in the 3-month follow-up, with no significant differences between the phenol and radiofrequency groups (Table 4). This suggests that both techniques are equally effective in controlling chronic knee pain.

The data also reveal that the majority of patients reported a reduction in the use of analgesics and a high level of satisfaction with genicular ablation (Table 2). Only a small percentage of patients experienced adverse effects, which is consistent with the existing literature that considers the procedure safe.

Knowledge of the anatomy of the genicular nerves is essential for performing ablation procedures. These structures are responsible for the sensory innervation of the knee joint capsule and primarily originate from branches of the femoral nerve. They are found in close relation to the epiphysis of the femur and tibia, and the use of radiological imaging to guide ablation procedures provides a robust foundation for safe and effective clinical practice.⁹

							0.209	<0.001	0.666
		6.6 ± 2.2	6.9 ± 3.1	9.1 ± 1	6.1 ± 3.3	5.7 ± 2.8			
		7 (5; 8)	7 (5; 10)	9 (8; 10)	6.5 (3.8; 9)	6 (4.3; 7.3)			
TUG							0.492	0.016	0.512
Average ± SD	17.5 ± 9.3	14.7 ± 5.6	15.4 ± 13.3	17.3 ± 8.2	13.3 ± 5.1	12.5 ± 4.1			
median (p25; p75)	14 (11.8; 17.4)	14 (11.5; 15.9)	12.4 (9.7; 16.7)	14.3 (11.6; 18.7)	10.9 (9.6; 16.9)	10.6 (9.4; 15.5)			
TLS30S							0.286	<0.001	0.375
Average ± SD	7 ± 2.3	8.6 ± 2.3	8.9 ± 2.9	6.3 ± 1.9	7.5 ± 2.6	8.7 ± 1.8			
median (p25; p75)	7 (5; 9)	8 (7; 9.8)	8.5 (6; 11)	6 (5; 8)	7 (6; 9.3)	8 (7; 10)			
KOOS General							0.520	<0.001	0.206
Average ± SD	25.8 ± 17	39.9 ± 20.2	41.6 ± 18.1	24.3 ± 12.5	41.6 ± 19.8	48.8 ± 20.6			
median (p25; p75)	24 (10.5; 41)	35 (27; 59)	43.5 (29.3; 53.5)	27 (10; 35)	42 (30; 52)	45 (33.8; 67.3)			
KOOS Symptoms+Rigidity*							0.834	<0.001	0.182
Average ± SD	38.7 ± 26.7	54.4 ± 27.6	54.6 ± 22.7	29.8 ± 13.1	54.9 ± 22.5	61.3 ± 23.9			
median (p25; p75)	50 (9; 61)	53.5 (33; 75)	54 (44.3; 73)	32 (20.3; 40.8)	61 (39.5; 68)	61 (44.3; 86.8)			
KOOS Pain*							0.463	<0.001	0.180
Average ± SD	24.9 ± 16.3	43.9 ± 24.1	46.1 ± 21.2	27.4 ± 16.6	44.9 ± 21.4	55.4 ± 24.6			
median (p25; p75)	28 (8; 40.5)	44 (26; 62.5)	45.5 (31; 64)	34.5 (7.5; 39.8)	47 (32; 57)	52.5 (37.5; 74.3)			
KOOS Daily Activity*							0.727	<0.001	0.462
Average ± SD	29.1 ± 20.8	44.6 ± 20.8	47.4 ± 21.7	29 ± 17.7	43.7 ± 25.9	54.6 ± 24			
median (p25; p75)	28 (11; 47)	35 (30.5; 61.5)	48 (32.8; 59.3)	33.5 (11.5; 41.8)	38 (29.5; 65.5)	45 (38; 80.5)			
KOOS Sports and Leisure Activity*							0.964	0.052	0.708
Average ± SD	6.8 ± 9.2	15.6 ± 20	14.6 ± 13.2	4.2 ± 5.8	18.5 ± 21.7	19.4 ± 14.7			
median (p25; p75)	0 (0; 10)	5 (0; 25)	12.5 (3.8; 21.3)	0 (0; 6.3)	15 (0; 27.5)	22.5 (3.8; 30)			
KOOS Quality of Life*							0.819	0.053	0.447
Average ± SD	13.4 ± 10.6	18.5 ± 16	19.3 ± 12.8	10.6 ± 12.8	19.4 ± 20.5	26.1 ± 20			
median (p25; p75)	13 (0; 22)	13 (6; 31)	19 (11.3; 25)	6 (0; 13.5)	19 (3; 31)	25 (15.8; 32.8)			

EEG with normal distribution and identity bond function. assuming matrix of correlations AR(1) between the moments; * Same model with logarithmic bond function.

Table 4. Results of multiple comparisons of scores that differed between assessment time points.

Variable	Comparison		Average difference	Standard error	p	IC (95%)	
						Lower	Superior
ECN	Pre/Initial -	1 month	2.86	0.48	<0.001	1.72	4.00
	Pre/Initial -	3 months	2.92	0.56	<0.001	1.59	4.26
	1 month -	3 months	0.06	0.50	>0.999	-1.12	1.25
TUG	Pre/Initial -	1 month	3.78	1.33	0.013	0.60	6.97
	Pre/Initial -	3 months	3.56	1.69	0.106	-0.49	7.61
	1 month -	3 months	-0.23	1.38	>0.999	-3.54	3.09
TLS30S	Pre/Initial -	1 month	-1.44	0.36	<0.001	-2.30	-0.59
	Pre/Initial -	3 months	-2.00	0.46	<0.001	-3.10	-0.90
	1 month -	3 months	-0.55	0.37	0.399	-1.43	0.33
KOOS General	Pre/Initial -	1 month	-15.57	2.54	<0.001	-21.66	-9.48
	Pre/Initial -	3 months	-18.89	3.32	<0.001	-26.84	-10.94
	1 month -	3 months	-3.32	2.66	0.638	-9.69	3.06
KOOS Symptoms+Rigidity	Pre/Initial -	1 month	-20.48	3.53	<0.001	-28.93	-12.04
	Pre/Initial -	3 months	-22.81	4.56	<0.001	-33.73	-11.89
	1 month -	3 months	-2.33	3.68	>0.999	-11.13	6.48
KOOS Pain	Pre/Initial -	1 month	-18.35	2.78	<0.001	-24.99	-11.70
	Pre/Initial -	3 months	-22.66	3.75	<0.001	-31.64	-13.67
	1 month -	3 months	-4.31	2.97	0.442	-11.43	2.81
KOOS Daily Activity	Pre/Initial -	1 month	-15.02	3.73	<0.001	-23.96	-6.08
	Pre/Initial -	3 months	-20.64	4.72	<0.001	-31.93	-9.35

tion method can be a more accessible e settings, given its greater cost-complexity. The radiofrequency procedure require more technology and training.⁴

Regarding the surgical treatment of severe knee osteoarthritis, arthroplasty is considered the gold standard. However, it presents higher risks, costs, and potential complications, along with a slower recovery. In this context, minimally invasive procedures such as genicular nerve ablation can be considered safe and effective, representing an important alternative for patients who are at prohibitive risk for arthroplasty or as a method for pain control and functional improvement in the medium term.¹¹

Although the results are promising, it is important to recognize the study's limitations. The relatively small sample size of 35 patients

effectiveness in the medium and long term. Additionally, the absence of a control group that did not receive interventional treatment may limit the ability to generalize the results.

CONCLUSION

Genicular nerve ablation in patients with severe knee osteoarthritis can provide lasting improvements in pain, function, and quality of life. Both methods, phenol and radiofrequency, were equally effective. Acknowledgements:

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REFERENCES